

Jaypee University of Engineering and Technology, Guna
B.Tech. CSE, Semester V
Advanced Programming Lab - II (CS215/ 18B17CI573)
Lab Exercise 9

Design and implement a **menu-driven Java program** to simulate a **Restaurant Food Ordering System** using **interfaces**, **arrays**, and **runtime polymorphism**. The program should allow users to select a restaurant, view its menu, place an order for multiple items, generate a bill, and confirm the order - all within a loop until the user chooses to exit.

Define an interface named **Restaurant** that declares the following methods:

- **void showMenu():** Displays the list of available food items and their prices.
- **Object[][] placeOrder():** Allows the customer to select multiple items, specify quantities, and returns a **2D array** of order details, where each row contains:

[Item Name, Quantity, Price].

- **double calculateBill(Object[][] orderDetails):** Accepts the 2D array returned by **placeOrder()** and computes the total bill amount for the customer's order.

Implement the **Restaurant** interface in the following two classes:

- **Class Dominos**
 - Displays its own customized menu.
 - Implements logic for placing orders and returning order details in a 2D array.
 - Implements bill calculation by summing up the total of all selected items.
- **Class KFC**
 - Displays its own customized menu
 - Allows users to select multiple items and specify their quantities.
 - Returns order details as a 2D array and computes the total bill accordingly.

Main Class: FoodOrderingSystem

Display a **main menu** with the following options:

FOOD ORDERING SYSTEM

1. Order from Dominos

2. Order from KFC

3. Exit

Enter your choice:

Based on the user's choice:

- Display the restaurant's menu using **showMenu()**.
- Call the **placeOrder()** method to accept order details and store them in a 2D array.
- Pass this array to **calculateBill()** to compute the total bill.
- Display the generated bill to the user.
- Ask for confirmation - whether the user wants to **confirm** or **cancel** the order.
- If confirmed, display an order success message.
- If canceled, discard the order and return to the main menu.
- Ensure that after every operation, the main menu reappears.
- The program should continue running until the user chooses the **Exit** option.

Additional Requirements:

- Use **only arrays** for managing menus and order details (avoid **ArrayList**, **Map**, or **Collections**).
- Demonstrate **runtime polymorphism** by invoking methods using a **Restaurant** reference variable.